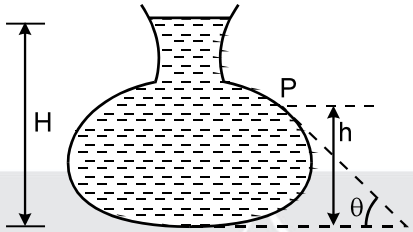
SYLLABUS : FLUID MECHANICS

1. Figure here shows the vertical cross-section of a vessel filled with a liquid of density ρ. The normal thrust per unit area on the walls of the vessel at point. P, as shown, will be



(A) $h\rho g$ (B) $H\rho g$ (C) $(H - h)\rho g$ (D) $(H - h)\rho g \cos\theta$

2. A tank with length 10 m, breadth 8 m and depth 6m is filled with water to the top. If $g = 10ms^{-2}$ and density of water is 1000 kg m^{-3}, then the thrust on the bottom is (neglect atmospheric pressure)

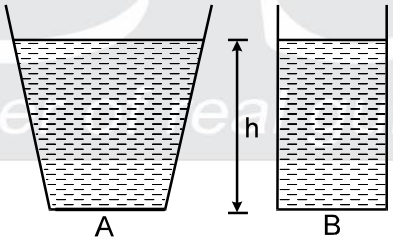
(A) $6 \times 1000 \times 10 \times 80$ N (B) $3 \times 1000 \times 10 \times 48$ N

(C) $3 \times 1000 \times 10 \times 60$ N (D) $3 \times 1000 \times 10 \times 80$ N

3. In a hydraulic lift, used at a service station the radius of the large and small piston are in the ratio of 20 : 1. What weight placed on the small piston will be sufficient to lift a car of mass 1500 kg ?

(A) 3.75 kg (B) 37.5 kg (C) 7.5 kg (D) 75 kg.

4. Two vessels A and B of different shapes have the same base area and are filled with water up to the same height h (see figure). The force exerted by water on the base is F_A for vessel A and F_B for vessel B. The respective weights of the water filled in vessels are W_A and W_B. Then



(A) $F_A > F_B$; $W_A > W_B$ (B) $F_A = F_B$; $W_A > W_B$

(C) $F_A = F_B$; $W_A < W_B$ (D) $F_A > F_B$; $W_A = W_B$

5. The density of ice is x gm/cc and that of water is y gm/cc. What is the change in volume in cc, when m gm of ice melts?

(A) M (y – x) (B) (y – x)/m (C) mxy (x – y) (D) m (1/y – 1/x)

6. The reading of a spring balance when a block is suspended from it in air is 60 newton. This reading is changed to 40 newton when the block is fully submerged in water. The specific gravity of the block must be therefore :

(A) 3 (B) 2 (C) 6 (D) 3/2

7. A block of volume V and of density σ_b is placed in liquid of density $\sigma_l(\sigma_l > \sigma_b)$, then block is moved upward upto a height h and it is still in liquid. The increase in gravitational potential energy of the block is :

(A) $\sigma_b Vgh$ (B) $(\sigma_b + \sigma_l)Vgh$ (C) $(\sigma_b - \sigma_l)Vgh$ (D) none of these

8. A block of steel of size 5 cm × 5 cm × 5 cm is weighed in water. If the relative density of steel is 7. Its apparent weight is :

(A) 6 × 5 × 5 × 5 gf (B) 4 × 4 × 4 × 7 gf (C) 5 × 5 × 5 × 7 gf (D) 4 × 4 × 4 × 6 gf

9. A metallic sphere floats in an immiscible mixture of water ($\rho_w = 10^3$ kg/m^3) and a liquid ($\rho_L = 13.5 \times 10^3$) with (1/5)th portion by volume in the liquid and remaining in water. The density of the metal is :

(A) 4.5×10^3 kg/m^3 (B) 4.0×10^3 kg/m^3 (C) 3.5×10^3 kg/m^3 (D) 1.9×10^3 kg/m^3

10. Two bodies are in equilibrium when suspended in water from the arms of a balance. The mass of one body is 36 g and its density is 9 g/cc. If the mass of the other is 48 g, its density in g/cc is :

(A) 4/3 (B) 3/2 (C) 3 (D) 5

11. In order that a floating object be in a stable equilibrium, its centre of buoyancy should be

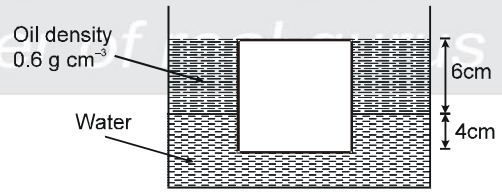
(A) vertically above its centre of gravity

(B) vertically below its centre of gravity

(C) horizontally in line with its centre of gravity

(D) may be anywhere

12. A cubical block of wood 10 cm on a side, floats at the interface of oil and water as shown in figure. The density of oil is 0.6 g cm^{-3} and density of water is 1 g cm^{-3}. The mass of the block is

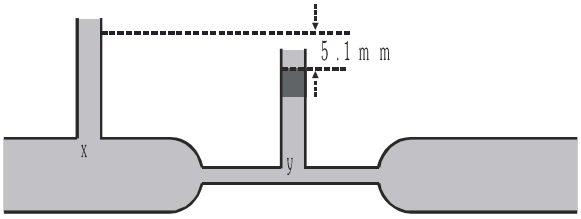


(A) 706 g (B) 607 g (C) 760 g (D) 670 g

13. A cube of wood supporting a 200 gm mass just floats in water. When the mass is removed the cube rises by 2 cm at equilibrium. Find side of the cube.

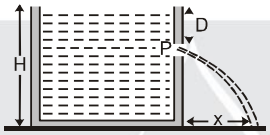
(A) 5 cm (B) 10 cm (C) 20 cm (D) 30 cm

14. The diagram (figure) shows a venturimeter, through which water is flowing. The speed of water at X is 2 cm/s. The speed of water at Y (taking g = 1000 cm/s^2) is :–



(A) 23 cm/s (B) 32 cm/s (C) 101 cm/s (D) 1024 cm/s

15. A tank is filled with water up to height H. Water is allowed to come out of a hole P in one of the walls at a depth D below the surface of water as shown in the figure. Express the horizontal distance x in terms of H and D :

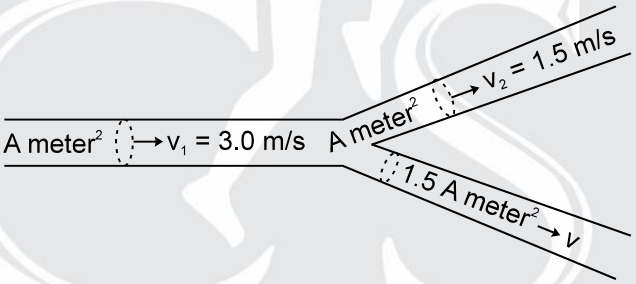


(A) $x = \sqrt{D(H-D)}$ (B) $x = \sqrt{\frac{D(H-D)}{2}}$ (C) $x = 2\sqrt{D(H-D)}$ (D) $x = 4\sqrt{D(H-D)}$

16. A fixed cylindrical vessel is filled with water up to height H. A hole is bored in the wall at a depth h from the free surface of water. For maximum horizontal range h is equal to :

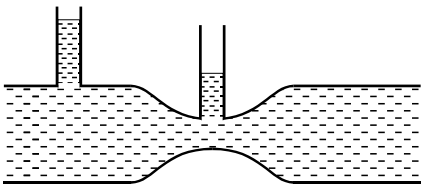
(A) H (B) 3H/4 (C) H/2 (D) H/4

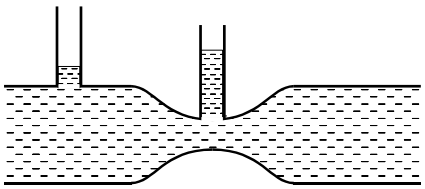
17. An incompressible liquid flows through a horizontal tube as shown in the figure. Then the velocity 'v' of the fluid is :

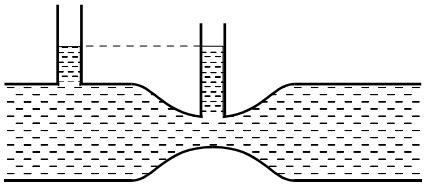


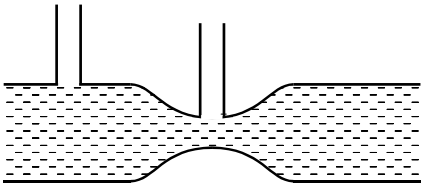
(A) 3.0 m/s (B) 1.5 m/s (C) 1.0 m/s (D) 2.25 m/s

18. For a fluid which is flowing steadily in a horizontal tube as shown in the figure, the level in the vertical tubes is best represented by

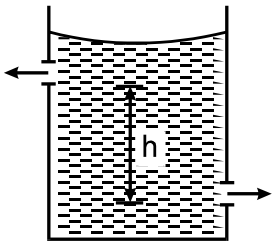
(A) 

(B) 

(C) 

(D) 

19. There are two identical small holes on the opposite sides of a tank containing a liquid. The tank is open at the top. The difference in height of the two holes is h as shown in the figure. As the liquid comes out of the two holes, the tank will experience a net horizontal force proportional to:



(A) $h^{1/2}$ (B) h (C) $h^{3/2}$ (D) h^2

20. A cylindrical tank of height 0.4 m is open at the top and has a diameter 0.16 m. Water is filled in it up to a height of 0.16 m. How long it will take to empty the tank through a hole of radius 5×10^{-3} m at its bottom ?

(A) 46.26 sec. (B) 4.6 sec. (C) 462.6 sec. (D) 0.46 sec.

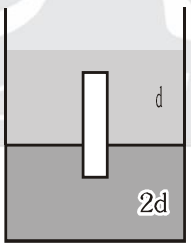
21. A large cylindrical vessel contains water to a height of 10m. It is found that the thrust acting on the curved surface is equal to that at the bottom. If atmospheric pressure can support a water column of 10m, the radius of the vessel is

(A) 10 m (B) 15m (C) 5m (D) 25m

22. A jet of water of cross-sectional area A hits a plate normally with velocity v. the plate is moving in the direction of the jet with velocity V. therefore, the force exerted on the plate is proportional to

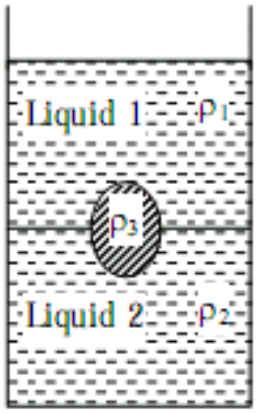
(A) v (B) v^2 (C) (v-V) (D) $(v-V)^2$

23. A homogeneous solid cylinder of length L (L < H/2), cross–sectional area A/5 is immersed such that it floats with its axis vertical at the liquid–liquid interface with length L/4 in the denser liquid as shown in the figure. The lower density liquid is open to atmosphere having pressure P_0. Then, density D of solid is given by :–



(A) $\frac{5}{4}d$ (B) $\frac{4}{5}d$ (C) 4d (D) $\frac{d}{5}$

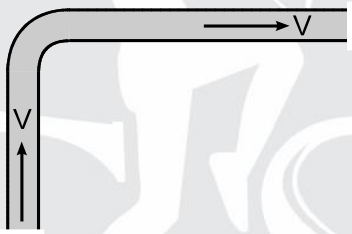
24. A jar is filled with two non-mixing liquids 1 and 2 having densities ρ_1 and ρ_2, respectively. A solid ball, made of a material of density ρ_3, is dropped in the jar. It comes to equilibrium in the position shown in the figure.



Which of the following is true for ρ_1, ρ_2 and ρ_3 ?

(A) $\rho_1 > \rho_3 > \rho_2$ (B) $\rho_1 < \rho_2 < \rho_3$ (C) $\rho_1 < \rho_3 < \rho_2$ (D) $\rho_3 < \rho_1 < \rho_2$

25. A fire hydrant (as shown in the figure) delivers water of density ρ at a volume rate L. The water travels vertically upward through the hydrant and then does 90^0 turn to emerge horizontally at speed V. The pipe and nozzle have uniform cross-section throughout. The force exerted by the water on the corner of the hydrant is



(A) ρVL (B) zero (C) $2\rho VL$ (D) $\sqrt{2}\rho VL$

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ANSWER KEY

1.	(C)	**2.**	(A)	**3.**	(A)	**4.**	(B)	**5.**	(D)
6.	(A)	**7.**	(A)	**8.**	(A)	**9.**	(C)	**10.**	(C)
11.	(A)	**12.**	(C)	**13.**	(B)	**14.**	(B)	**15.**	(C)
16.	(C)	**17.**	(C)	**18.**	(A)	**19.**	(B)	**20.**	(A)
21.	(B)	**22.**	(D)	**23.**	(A)	**24.**	(C)	**25.**	(D)